

Appendix

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June 2026

1 Optimization method and hypothesis

In standard OLS, because the squared loss function creates a smooth parabola, taking the derivative and setting it to zero yields a clean, closed-form algebraic solution: $\hat{\beta} = (X^\top X)^{-1} X^\top Y$. However, the absolute value in our Check Function creates a sharp "V" shape at zero. This non-differentiable corner means we cannot use standard First-Order Conditions. Instead, the optimization must be transformed into a Linear Programming (LP) problem.

To eliminate the absolute piecewise nature of the function, we split every residual into two strictly non-negative directional buckets:

- u_i^+ (Under-prediction bucket): The magnitude by which the actual value exceeded the prediction.
- u_i^- (Over-prediction bucket): The magnitude by which the prediction exceeded the actual value.

Every residual can now be expressed as a linear combination:

$$u_i = u_i^+ - u_i^-$$

By substituting this into our regression equation ($y_i - x_i^\top \beta = u_i^+ - u_i^-$) and rearranging, we define the strict structural constraints for the optimization algorithm:

$$x_i^\top \beta + u_i^+ - u_i^- = y_i$$

Applying this transformation completely removes the absolute value operator from the objective function. The complex minimization problem is elegantly simplified into a standard primal Linear Programming format:

Minimize:

$$\min_{\beta, u^+, u^-} \sum_{i=1}^n [\tau u_i^+ + (1 - \tau) u_i^-]$$

Subject to:

1. $x_i^\top \beta + u_i^+ - u_i^- = y_i$ (The LP Constraint)
2. $u_i^+, u_i^- \geq 0$ (Positivity Constraint)

By formulating the math this way, statistical software can solve the regression using highly efficient LP algorithms, primarily the Barrodale-Roberts Algorithm (Modified Simplex) for smaller datasets, which searches for the minimum along the geometric edges of the data intersections, or Frisch-Newton Interior Point Methods for large datasets, which cut directly through the feasible space using a logarithmic barrier to ensure positivity.

When adapting this methodology for predictive econometrics (such as the Growth-at-Risk or Oil-at-Risk framework), the model must predict future values h -steps ahead (y_{t+h}) using only current conditioning variables (x_t). Unlike iterated forecasting (e.g., standard ARMA models)—which predicts y_{t+1} and chains errors forward—Direct Forecasting shifts the target variable backward by h periods. The training data must stop at $T - h$ to allow the computation of actual forecast errors. The final objective function for a direct forecasting quantile regression becomes:

$$\hat{\beta}_\tau = \arg \min_{\beta_\tau \in R^k} \sum_{t=1}^{T-h} \rho_\tau(y_{t+h} - x_t \beta_\tau)$$

Once optimized, the resulting $\hat{\beta}_\tau$ provides a consistent linear estimator of the conditional quantile function, allowing us to map out the explicit vulnerabilities of y_{t+h} to shocks in x_t without the compounding error risks inherent in iterated models.

For $\hat{\beta}_\tau$ to be a consistent estimator, the model relies on several core hypotheses:

- **Linearity of Conditional Quantiles:** The conditional quantile is assumed to have a linear relationship with the regressors, such that $Q_\tau(Y|X) = X^\top \beta(\tau)$.
- **Strict Exogeneity:** Instead of assuming the expected value of the errors is zero, we assume the conditional quantile of the error term, given X , is zero: $Q_\tau(u|X) = 0$.
- **No Multicollinearity:** The matrix $E[XX^\top]$ must be finite, positive-definite, and invertible.
- **Positive Density:** The conditional density function of the error term around zero must be strictly positive and finite ($0 < f_{u|X}(0|X) < \infty$). Because precision in quantile regression depends on the mass of data points falling exactly on the regression line, a lack of density at the evaluated quantile causes the variance of the estimator to explode.

A Quantile Regression Diagnostics: Full Technical Development

This appendix presents the complete methodological foundation of each diagnostic test applied to the specification. For each test, the statistical framework, null hypothesis, test statistic, and asymptotic distribution are stated precisely. The numerical results referenced in the main text are reported in the accompanying tables.

A.1 Goodness-of-Fit: Koenker-Machado $R^1(\tau)$

Unlike Ordinary Least Squares, quantile regression does not minimise a quadratic objective and therefore possesses no natural coefficient of determination. The local goodness-of-fit measure introduced by Koenker and Machado (1999) resolves this by defining a pseudo- R^2 as the proportional reduction in the asymmetrically weighted sum of absolute errors relative to an intercept-only null model. Formally:

$$R^1(\tau) = 1 - \frac{\sum_{t=1}^T \rho_\tau(y_t - \mathbf{x}'_t \hat{\beta}(\tau))}{\sum_{t=1}^T \rho_\tau(y_t - \hat{\xi}_\tau)}, \quad (1)$$

where $\rho_\tau(u) = u[\tau - \mathbf{1}(u < 0)]$ is the check function defined in previous sections and $\hat{\xi}_\tau$ is the unconditional sample τ -quantile, which is the solution to the intercept-only problem. The metric is bounded in $[0, 1]$ and interpreted locally: a value of $R^1(\tau) = 0.10$ at $\tau = 0.05$ indicates that the regressors reduce the weighted objective by ten percent relative to the unconditional quantile at that tail. Koenker and Machado (1999) establish that the statistic is consistent and that, under standard regularity conditions, the value $T \cdot V(\tau) \cdot R^1(\tau)$ is asymptotically chi-squared with k degrees of freedom, where k is the number of slope parameters and $V(\tau)$ is a nuisance parameter that depends on the sparsity of the error distribution at τ .

A.2 Wald Test for Equality of Coefficients across Quantiles

A foundational identifying assumption of the Oil-at-Risk framework is that the effect of geopolitical risk on the return distribution is heterogeneous across quantiles. If the slope coefficients are identical for all τ , the distribution shifts in parallel and quantile regression reduces to a reparameterisation of OLS. The Wald test proposed by Koenker and Bassett (1982) formalises this by testing the null hypothesis of coefficient constancy across m quantiles jointly.

Let $\hat{\zeta} = (\hat{\beta}(\tau_1)', \dots, \hat{\beta}(\tau_m)')'$ be the $mp \times 1$ stacked parameter vector across all m estimated quantiles. A full-rank $q \times mp$ restriction matrix R is constructed

to extract the relevant slope coefficients and impose equality across quantile pairs, so that the null hypothesis reads $H_0 : R\hat{\zeta} = r$, where r is the $q \times 1$ vector of hypothesised values (typically zero for difference restrictions). The Wald statistic is:

$$T_n = n(R\hat{\zeta} - r)^\top \left[R \hat{V}_n R^\top \right]^{-1} (R\hat{\zeta} - r) d\chi_q^2, \quad (2)$$

where $\hat{V}_n$ is the joint asymptotic covariance matrix of $\hat{\zeta}$. Because the parameters are estimated from the same time series, their errors are correlated across quantiles, and the off-diagonal blocks of $\hat{V}_n$ are non-zero. Analytical sparsity estimators are insufficient in the presence of temporal dependence; accordingly, $\hat{V}_n$ is constructed via a moving block bootstrap, which resamples contiguous blocks of observations of length l chosen to capture the autocorrelation horizon of the data, re-estimating the full quantile system at each bootstrap replication.

A.3 Quantile Crossing Inspection

A necessary condition for any valid conditional distribution function is that the conditional quantile function be monotonically non-decreasing in τ . Formally, for any two quantile levels $\tau_1 < \tau_2$ and any observation t :

$$\hat{q}_t(\tau_1) \leq \hat{q}_t(\tau_2) \quad \forall t \in \{1, \dots, T\}. \quad (3)$$

Violations of this condition, known as quantile crossings, arise when the estimated linear quantile planes intersect in the covariate space, typically as a consequence of over-parameterisation, collinear regressors, or an insufficient number of observations relative to the complexity of the model. Crossings are detected by computing, for each observation t , the ordered sequence of fitted values across all targeted quantiles and checking for any inversion. The proportion of crossing observations and the magnitude of the maximal inversion are reported as informal diagnostics; a specification with systematic crossings in the interior of the covariate support is treated as misspecified.

A.4 Stationarity: Quantile Autoregressive Unit Root Test

Standard unit root tests evaluate the null of a unit root against the alternative of global mean reversion, implicitly assuming that the autoregressive parameter is homogeneous across the distribution. Koenker and Xiao (2004) relax this assumption by permitting quantile-specific autoregressive dynamics, allowing the series to be stationary at some quantiles while exhibiting near-unit-root behaviour at others. The test is implemented via the auxiliary error correction regression:

$$\Delta y_t = \mu(\tau) + \rho(\tau) y_{t-1} + \sum_{j=1}^p \gamma_j(\tau) \Delta y_{t-j} + \epsilon_t(\tau), \quad (4)$$

where the lag order p is selected by standard information criteria applied to the ADF pre-regression. The null hypothesis $H_0 : \rho(\tau) = 0$ posits a quantile-specific unit root and is tested against the one-sided alternative $H_1 : \rho(\tau) < 0$. The test statistic $t_{QAR}(\tau) = \hat{\rho}(\tau)/\widehat{se}(\hat{\rho}(\tau))$ is obtained from the quantile objective function and does not follow a standard normal distribution under the null. Koenker and Xiao (2004) establish that its limiting distribution is:

$$t_{QAR}(\tau) d \frac{f(F^{-1}(\tau))}{\sqrt{\tau(1-\tau)}} \cdot \frac{\int_0^1 W(s) dW(s)}{\left(\int_0^1 W(s)^2 ds\right)^{1/2}}, \quad (5)$$

where $W(s)$ is a standard Brownian motion on $[0, 1]$, f is the marginal density of the innovations, and $F^{-1}(\tau)$ is the corresponding population quantile. The right-hand side is the standard Dickey-Fuller functional scaled by the sparsity factor $f(F^{-1}(\tau))/\sqrt{\tau(1-\tau)}$, which varies with τ and renders standard Dickey-Fuller critical values inapplicable. Quantile-specific critical values are obtained by simulation following Koenker and Xiao (2004).

A.5 Dynamic Quantile Test

The Dynamic Quantile test of Engle and Manganelli (2004) evaluates whether the sequence of quantile violations constitutes a martingale difference sequence. The centred hit sequence is defined as:

$$Hit_t(\tau) = \mathbf{1}(y_t < \hat{q}_t(\tau)) - \tau, \quad (6)$$

which is mean zero by construction under correct specification. The null hypothesis is $H_0 : E[Hit_t(\tau) \mid \Omega_{t-1}] = 0$, meaning the violation process is orthogonal to all past information. To test this, the hits are projected via OLS onto an instrument matrix $\mathbf{Z}_t$ comprising a constant, L lags of $Hit_t(\tau)$, and lagged fitted quantiles $\hat{q}_{t-1}(\tau)$:

$$Hit_t(\tau) = \mathbf{z}_t' \delta + \eta_t. \quad (7)$$

Engle and Manganelli (2004) show that under the null, the Wald form of the test statistic,

$$DQ(\tau) = \frac{\mathbf{Hit}' \mathbf{Z}(\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{Hit}}{\tau(1-\tau)} d\chi_q^2, \quad (8)$$

is asymptotically chi-squared with q degrees of freedom equal to the column dimension of $\mathbf{Z}$. The equivalent $N \cdot R^2$ form of the LM statistic converges to the same distribution asymptotically and is reported for comparability. The test is applied separately at each targeted quantile; a joint test across all quantiles is constructed by stacking the instrument matrices and adjusting the degrees of freedom accordingly.

A.6 Conditional Heteroskedasticity Test

The ARCH-LM test of Engle (1982), applied to the quantile residuals $\hat{\epsilon}_t(\tau) = y_t - \mathbf{x}_t' \hat{\beta}(\tau)$, evaluates whether the squared residuals exhibit autoregressive conditional heteroskedasticity. The null hypothesis is:

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0 \quad \text{in} \quad \hat{\epsilon}_t^2(\tau) = \alpha_0 + \sum_{i=1}^p \alpha_i \hat{\epsilon}_{t-i}^2(\tau) + \nu_t, \quad (9)$$

which corresponds to $E[\hat{\epsilon}_t^2(\tau) \mid \Omega_{t-1}] = \sigma_\tau^2$, a constant conditional variance. The lag order is set to $p = 5$ to span a full trading week. The Lagrange multiplier statistic $LM = T \cdot R^2$, where R^2 is the coefficient of determination of the auxiliary regression is asymptotically χ_p^2 under the null. When applied within a quantile regression context, the test examines whether the chosen specification adequately captures the heteroskedastic structure of the return series at each quantile level, rather than merely absorbing it at the conditional mean. Failure to reject provides evidence that the estimated quantile coefficients are not contaminated by residual volatility clustering.

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Annex A

Out-of-Sample Forecast Evaluation Tests in Detail

This annex documents, in full, every test used to evaluate the direct-forecasting models of Section 2. For each test we give the intuition, the null and alternative hypotheses, the exact test statistic and its asymptotic distribution, the implementation choices made in `aux/diagnostics/direct_forecasting.py`, and the realised result for the three competing models. Two conventions hold throughout. First, a **hit** (or breach) is defined under a unified directional rule: a hit occurs whenever the realised value falls strictly below the forecast, $y_{t+h} < \hat{y}_{t+h|t}(\tau)$, so that for any quantile the model should be breached exactly $\tau \times 100\%$ of the time. Second, all coverage tests are run out-of-sample on a fixed 1,000-day rolling window, re-estimated at every forecast origin.

B Evaluation protocol: the rolling-window experiment

Every statistic below is computed on genuinely out-of-sample forecasts generated by a fixed-width rolling window. For each forecast origin t in the test span the model is trained on the 1,000 observations in the half-open interval $[t - 1000, t)$, a single h -step-ahead prediction of Y_{t+h} is produced, and the window then rolls forward one day, dropping its oldest observation. Because the realised target Y_{t+h} never enters the training window, there is no load-bearing look-ahead. *(One mild caveat is documented in the code: for $h > 1$ the training*

rows near the window edge carry shifted targets dated at or after t , the standard direct-forecasting convention, which introduces a small optimistic bias that is acknowledged rather than corrected.)

A single forecast error is never conclusive. The reported figures are therefore averages over hundreds of origins: 867 out-of-sample forecasts for the benchmark (2 January 2023 – 28 April 2026) and 519 for the CAViaR variants (4 January 2023 – 24 April 2026), the difference reflecting the extra rows consumed by constructing the breach regressors.

C The pinball loss and the generalization gap

In conditional-mean forecasting the mean-squared error is the rational default. It is, however, symmetric, and symmetry is precisely the wrong property for a quantile, where over- and under-prediction must be penalised by different amounts. The asymmetric **pinball (check) loss** supplies that asymmetry:

$$L_\tau(e) = \begin{cases} \tau & \text{if } e \geq 0, \\ (\tau - 1) & \text{if } e < 0, \end{cases} = e \cdot (\tau - \mathbf{1}\{e < 0\}), \quad (10)$$

with error $e = y_{t+h} - \hat{y}(\tau)$. The loss is always non-negative and is the element-wise object returned by `tick.loss.series`; its mean is the `pinball.loss` reported across the notebook.

C.1 Why the pinball loss and not any other

A scoring rule is **strictly proper** if it is minimised in expectation if and only if the issued forecast equals the true value of the targeted functional. The pinball loss is the unique (up to affine transformation) strictly proper scoring rule for a quantile. The practical consequence is strong: if model A has a lower out-of-sample pinball loss than model B at quantile τ , then A is provably a better estimator of that τ -quantile — the comparison needs no distributional assumption. This is what licenses the loss-based model ranking in Section 3.

C.2 The generalization gap

Evaluating the loss both in-sample (on the training window) and out-of-sample (on unseen data) across a dense grid of horizons $h = 1, \dots, H$ yields the **generalization gap**, the shaded area between the two loss curves. A small, stable gap indicates that the model captures structural signal; a gap that widens sharply with the horizon is the classic signature of overfitting — the regression fitting localised noise that does not survive into the test period. In our experiment the gap stays bounded for the benchmark across all horizons, whereas the CAViaR variants begin to separate beyond $h \approx 5$, consistent with their stale-information handicap.

C.3 Realised result

Averaging the rolling out-of-sample pinball loss over the test window and selecting, for each model and quantile, the loss-minimising horizon gives:

Model	$\tau = 0.05$	$\tau = 0.50$	$\tau = 0.95$
Standard QReg	$h=7 \rightarrow 0.5295$	$h=3 \rightarrow 1.6061$	$h=1 \rightarrow 0.5624$
CAViaR-indicator	$h=9 \rightarrow 0.6342$	$h=1 \rightarrow 1.6210$	$h=1 \rightarrow 0.5206$
CAViaR-severity	$h=5 \rightarrow 0.6860$	$h=1 \rightarrow 1.6457$	$h=1 \rightarrow 0.5432$

Table 1: Loss-minimising horizon per model and quantile (lower is better). The CAViaR variants win the 95% tail only at $h = 1$; the benchmark dominates the median and the lower tail at all economically relevant horizons.

D Fallout (severeness of breach) audit

A model can pass a frequency test yet still fail by suffering rare but enormous overshoots. The fallout audit measures not whether a breach occurred but *by how much* the boundary was overshoot. Fitting a lower ($\tau = 1\%$) and upper ($\tau = 99\%$) boundary and comparing them to the realised series, the fallout error is

$$E_t = \max(0, Q_{\text{low}} - y_t) + \max(0, y_t - Q_{\text{high}}), \quad (11)$$

which is zero on safe days and equal to the exceedance distance on breach days. The *severeness indicator* used in Model 3 is exactly this quantity, split into asymmetric upper/lower components and entered in absolute value so its coefficient measures a pure hit magnitude. For risk management a model with many tiny breaches strictly dominates one with few gigantic breaches, because the latter signals a structural failure to anticipate a shock. At $h = 2$ the share of test days kept inside the 1%–99% tunnel is **97.86%** (Standard), **97.26%** (CAViaR-i) and **97.26%** (CAViaR-s), with no isolated catastrophic overshoot in any model.

E Kupiec Proportion-of-Failures test (unconditional coverage)

Before worrying about dynamics we must answer the simplest question: does the 10th-percentile model actually leave 10% of the data below it? The Kupiec (1995) POF test checks whether the empirical hit frequency matches the nominal τ . It is *unconditional* because it judges frequency alone, ignoring timing. It is the backtesting analogue of the PIT check in distribution fitting, where each quantile’s density should integrate to its nominal level.

Let T be the number of out-of-sample forecasts and x the number of hits, with empirical rate $\hat{\pi} = x/T$. Since independent breaches are Bernoulli, the null

and alternative are

$$H_0 : E[I_{t+h}] = \tau \quad vs. \quad H_1 : E[I_{t+h}] \neq \tau. \quad (12)$$

The likelihood-ratio statistic contrasts the restricted likelihood (probability fixed at τ) with the unrestricted one (probability free at $\hat{\pi}$):

$$LR_{UC} = -2 \ln \frac{(1-\tau)^{T-x} \tau^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2. \quad (13)$$

By Wilks' theorem $-2 \ln \Lambda$ converges to a chi-squared variable whose degrees of freedom equal the difference in free parameters — here one (the unrestricted rate) minus zero (the fixed target). Because the hit count is binomial, the central limit theorem also supplies a normal-approximation confidence band around τ , $\tau \pm z\sqrt{\tau(1-\tau)/n}$, which is the funnel drawn in the coverage plots and the statistical basis of the Basel traffic-light VaR classification.

E.1 Realised result ($\tau = 0.95$, $h = 2$)

Model	Exp. hits	Act. hits	Emp. rate	LR_{UC}	p -value	Decision
Standard QReg	823.6/867	833	0.9608	2.285	0.131	Fail to reject
CAViaR-indicator	493.0/519	499	0.9615	1.554	0.213	Fail to reject
CAViaR-severity	493.0/519	499	0.9615	1.554	0.213	Fail to reject

Table 2: Kupiec POF results. All three fail to reject the null: their unconditional coverage is statistically correct. The empirical rates (96.1%) sit slightly above the 95% target but well inside the confidence band.

F Christoffersen independence test

Kupiec is blind to clustering: a model that breaches ten days in a row and then stays quiet can still post a perfect average rate. Christoffersen (1998) closes this gap by asking *when* breaches occur. The hit sequence is modelled as a first-order Markov chain, and we count the four transition types:

Symbol	Meaning
T_{00}	safe day followed by a safe day
T_{01}	safe day followed by a breach
T_{10}	breach followed by a safe day
T_{11}	breach followed by a breach

From these counts the state-conditional breach probabilities are

$$\pi_{01} = \frac{T_{01}}{T_{00} + T_{01}}, \quad \pi_{11} = \frac{T_{11}}{T_{10} + T_{11}}, \quad \pi = \frac{T_{01} + T_{11}}{T_{00} + T_{01} + T_{10} + T_{11}}. \quad (14)$$

If the model has properly absorbed shocks, today’s breach must be independent of yesterday’s, giving

$$H_0 : \pi_{01} = \pi_{11} = \pi \quad vs. \quad H_1 : \pi_{01} \neq \pi_{11}. \quad (15)$$

The restricted likelihood imposes a single common breach probability; the unrestricted one lets the two transition probabilities differ: $L_{\text{indep}} = (1-\pi)^{T_{00}+T_{10}} \pi^{T_{01}+T_{11}}$, $L_{\text{dep}} = (1-\pi_{01})^{T_{00}} \pi_{01}^{T_{01}} (1-\pi_{11})^{T_{10}} \pi_{11}^{T_{11}}$, $LR_{IND} = -2 \ln \frac{L_{\text{indep}}}{L_{\text{dep}}} \sim \chi_1^2$. The implementation guards every logarithm against $\log(0)$ and floors the statistic at zero to absorb floating-point noise, so the test is numerically stable even when a transition cell is empty.

G Christoffersen combined conditional-coverage test

Christoffersen’s decisive contribution is to test both properties simultaneously. Because the unconditional-coverage statistic (Kupiec) and the independence statistic are asymptotically independent, their sum is again chi-squared and the degrees of freedom add:

$$LR_{CC} = LR_{UC} + LR_{IND} \sim \chi_2^2 \quad (16)$$

A model passes only if its breaches arrive at the right *frequency* (the Kupiec component) *and* at independent *times* (the independence component). This is the single most demanding coverage criterion and the one used in the main text to certify the production model.

G.1 Realised result ($\tau = 0.95$, $h = 2$)

Model	Test window	N	LR_{CC} p -value	Decision
Standard QReg	2023-01-02 → 2026-04-28	867	0.136	Fail to reject
CAViaR-indicator	2023-01-04 → 2026-04-24	519	0.445	Fail to reject
CAViaR-severity	2023-01-04 → 2026-04-24	519	0.445	Fail to reject

Table 3: Christoffersen combined test. Every model fails to reject correct conditional coverage at the 5% level: breaches are both correctly frequent and serially independent. No breach clustering is detected in any specification.

H Diebold–Mariano test of equal predictive accuracy

The coverage tests certify each model alone; the Diebold–Mariano (1995) test compares two models head-to-head. Let $L_{1,t}$ and $L_{2,t}$ be the element-wise tick

losses of two competing forecasts and $d_t = L_{1,t} - L_{2,t}$ their differential. The null of equal accuracy is $H_0 : E[d_t] = 0$. The statistic standardises the mean differential by a heteroskedasticity- and autocorrelation-consistent (HAC) long-run variance:

$$DM = \frac{\bar{d}}{\sqrt{LRV/P}}, \quad LRV = \gamma_0 + 2 \sum_{k=1}^{h-1} \gamma_k \sim t_{P-1}. \quad (17)$$

Two implementation choices follow the theory exactly. The HAC bandwidth is set to $h - 1$, because the errors of an h -step direct forecast follow an $MA(h - 1)$ process and therefore have exactly $h - 1$ non-zero autocovariances; truncating the kernel there is efficient and unbiased. The p -value is taken from the Student- t distribution with $P - 1$ degrees of freedom rather than the normal, a small-sample correction. A negative, significant $\alpha = \bar{d}$ means the first model has the lower average loss.

H.1 Realised result ($\tau = 0.95$, $h = 2$)

Comparison	α	t -stat	p -value	Significance
Standard vs CAViaR-indicator	-0.00647	-0.874	0.382	none
Standard vs CAViaR-severity	-0.00620	-0.925	0.355	none
CAViaR-indicator vs CAViaR-severity	+0.00027	+0.158	0.875	none

Table 4: Pairwise Diebold–Mariano tests.

Companion error metrics: average tick loss 0.2516/0.2581/0.2578 and RMSE 4.500/4.520/4.464 for Standard, CAViaR-i and CAViaR-s respectively. No pairwise comparison is significant even at the 10% level — the three models are statistically indistinguishable in predictive accuracy, so parsimony decides in favour of the benchmark.

I Summary of evidence

Conclusion. All three specifications pass every coverage backtest at the 5% level. The Diebold–Mariano test finds no statistically significant accuracy difference among them. The standard quantile-regression direct forecast is therefore retained as the production model on the grounds of parsimony, the lowest tick loss, and the most stable generalisation across the severity-optimised horizons $h \in \{2, 12, 28, 63\}$.

Test	What it certifies	Standard	CAViaR-i	CAViaR-s
Pinball / generalization gap	Proper-scoring fit, no over-fit	Pass	Pass	Pass
Fallout audit	No catastrophic overshoot	97.86%	97.26%	97.26%
Kupiec POF	Correct breach frequency	Pass (0.131)	Pass (0.213)	Pass (0.213)
Christoffersen combined	Frequency + independence	Pass (0.136)	Pass (0.445)	Pass (0.445)
Diebold–Mariano	Relative accuracy	baseline	n.s.	n.s.

Table 5: Evidence summary across the three direct-forecasting models.